

Double-Flip Sequential CFET: Superior 3T Library Efficiency and PPA

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Abstract

Double-flip sequential CFET (sCFET) architecture enables split-gate devices and dense BM0 routing, supporting 3-track standard cells with high library efficiency. Hybrid orientation and asymmetric nanosheet widths in sCFET further optimize device-level PPA, offering scaling potential.

Introduction

Common-gate (CG) CFETs lack the independent gate control needed for transmission gates, while split-gate (SG) devices address this limitation [1,2]. To enable split-gate within realistic process margins, we propose a double-flip sCFET at the A7 node with asymmetric stacking and a local vertical bypass path (Fig.1d) [3-5]. Combined with dense BM0 routing, this enables efficient 3-track CFET standard cells. sCFET supports hybrid top-bottom device orientation and asymmetric sheet widths to enhance PPA and design flexibility.

Double-Flip Sequential Processing flow

CFET devices can be patterned using two approaches. In monolithic CFET (mCFET), the top and bottom devices are patterned in a single step using the same masks for Active and Gate. Sequential CFET instead patterns the two devices independently with dedicated masks. This increases the mask-count and extends the process flow (Table I) but enables greater design freedom and architectural flexibility.

The sCFET process flow is shown in Fig. 2 and starts from a bonded stack with embedded Middle Dielectric Isolation (eMDI) between top and bottom devices. First the top device is patterned from the front stopping on the eMDI up till the first (high-T) part of the RMG process (high-k and dipole but no WF-metals yet). Next, a temporary layer is bonded on the top, the wafer is flipped, and the bulk of the bottom wafer is removed. Bottom device is patterned from the backside through all high-temperature steps. The connection between bottom and top Gate through eMDI is a new step in sCFET requires an isolation etch inside the gate trench without attacking the devices, which is enabled by an additional BDI protection layer. It has been shown [9] that overlay between the Top and Bot devices is 3 nm. The top and bottom Device are finished with Nanosheet-like process with another Flip.

The sCFET scheme enables split gate and offers two key benefits: a two-step RMG for the top device, in which the second (metal-fill) step is performed at low thermal budget, enabling low-V_t for both tiers, and the sequential patterning of a single device at a time reduces etch complexity by avoiding high-aspect-ratio processes inherent to monolithic CFET.

Standard Cell Library Layout Efficiency

Impacts of architectural choices (i.e. common vs split-gate, dense backside BM0, MOL connectivity) can be obscured if only evaluated at a single cell. Therefore, several CFET libraries (Fig.1) have been designed to evaluate routability, area scaling and library efficiency. Cell area breakdown and core area scaling projections are shown in Table II and Fig.3 for each CFET architecture. Library layout efficiency (η_{layout}) is defined as follows:

$$\eta_{\text{layout}} = \text{scaling}_{\text{Core Area}} / \text{scaling}_{\text{Unit Area}}, \quad (1)$$

where *Core Area* and *Unit Area* scaling are normalized to the reference N2 6T Nanosheet node with backside power [10]. *Unit area* is Contacted Poly Pitch (CPP) × Cell Height (CH). *Core area* is the projected total cell area of an ARM 64-bit HD core with ~500K-700K cells (~10%-7% flop rate) without any SRAM macros, targeted at lvt, tt, 0.7 V, 25 °C corner.

Double-Row CFET (DR) [6, 7] cells show significant area overhead caused by intrinsic layout limitations. To overcome

DR layout inefficiencies, half-height DR (*hDR*) architecture was proposed in [8], and recent PPA results have been presented in [7]. Next, *hDR* with dense-pitch backside BM0 (*hDR denseBM0*) and *hDR split-gate* comparably improve cell area and layout efficiency. Finally, sCFET combines both dense backside and split-gate capabilities within feasible process margins. Fig.3 shows that sCFET achieves the lowest core area (-23%) with the highest layout efficiency (98%).

Power/Performance Evaluation

Gate capacitance evaluation: In the CFET structure with split gate enabled by double-flip processing, the taller backside gate increases parasitic gate capacitance as shown in Fig.4. This added capacitance leads to a ~10% RO frequency reduction. Sensitivity analysis in Fig.5 indicates that reducing the gate, MD, and BMD heights can shrink the frequency gap between common-gate and split-gate devices from 10% to ~4%.

VGG thickness in Split gate: Complex cells (e.g., MUX) contain both connected gates of the same polarity and isolated gates of opposite polarity as shown in Fig.6. Connected gates benefit from a smaller VGG to minimize fringing capacitance, while isolated opposite-polarity gates prefer a larger VGG to reduce gate coupling. Selecting the optimal VGG requires balancing these trade-offs with process complexity.

sCFET with hybrid orientation (HOT): Independent NMOS/PMOS crystal-orientation optimization in sCFET delivers 19-27% Ion gains for stressed channels (Fig.7), boosting sCFET performance from 10% below mCFET CG to 9% above. Together with gate-capacitance tuning, the sCFET RO speed improves by ~17% over mCFET CG. These gains increase power density, but reducing the active sheet width to 14 nm lowers it and enables a smaller standard-cell height (Fig.8-9). At 14 nm sheet width, the CH can be scaled to 54 nm. sCFET also uses only 3 M0 tracks, 0.5 fewer than mCFET, while maintaining an 18 nm M0 pitch at 54 nm CH.

Asymmetrical sheet width: In the sCFET architecture, with the VMM only bypassing the bottom device, high library efficiency is maintained, allowing the top device to use a wider sheet. This can be used to balance rise/fall delays by strengthening the weaker device. As shown in Fig.9, widening the NMOS sheet (e.g., 14 nm → 18 nm) restores its drive current and re-equalizes rise/fall delays. In complex cells such as NAND, series-connected NMOS devices amplify the impact of the weaker transistor. The stacked pull-down path has higher effective resistance. Enlarging the NMOS sheet width reduces this resistance, giving greater frequency benefits in NAND and other stacked-NMOS structures.

Conclusion

Double-flip sCFET enables split-gate architecture, together with dense BM0 routing and the VMM scheme, it supports a 3T device without library efficiency penalty. The 3T architecture can be further enhanced through capacitance tuning and HOT optimization. The combination of 3T design and HOT creates additional scaling potential. The asymmetrical sCFET structure enables hybrid sheet-width optimization, allowing independent NMOS/PMOS tuning for balanced electrical performance and improved delay in complex stacked cells such as NAND.

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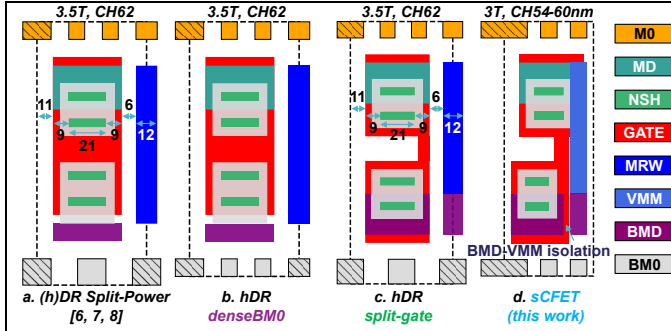


Fig. 1. Cross-sectional views of a. (h)DR mCFET [6-8], b. with dense BM0 tracks (i.e. same M0 pitch), c. with split-gate, d. sCFET with split-gate and dense BM0. sCFET CPP, CH and M0 pitch/cd are 42, 54-60 and 18/10 nm, respectively. Power and ground rails are ~3-cd wide. Both top and bottom FETs have symmetric routing resources at front/backside, i.e. 2 signal routing tracks per n/pFET, respectively. Vertical local interconnect is realized by VMM via (that can bypass bottom device). All regions are encapsulated within cell boundaries, hence sCFET doesn't cause any resource conflicts/constraints as opposed to hDR mCFET architectures with shared MRW at cell boundary.

TABLE I. Flow comparison of different processing schemes. The comparison considers a FEOL flow from Backside-M0 to Frontside-M0, using one VT for nMOS and pMOS each (2 in total). No IO-devices, diodes etc. are considered. The DR and hDR flows are identical in process and differ only in layout. Adding the denseBM0 increases the mask-count by 17% and the op-count by 11%, whereas adding split-gate increases the mask-count also by 17% and the op-count by 16%. sCFET has the combination of both elements, this leads to a mask-count increase by 55% and op-count by 37%.

	(h)DR [6,7,8]	hDR denseBM0	hDR split-gate	sCFET This work
Integration	Monolithic	Monolithic	Monolithic	Sequential
EUV layers	18	21	21	28
#operations	216	241	251	300
BM0 pitch (nm)	31	18	31	18
Hybrid orient.	No	No	No	Yes
Split gate	No	No	Yes	Yes

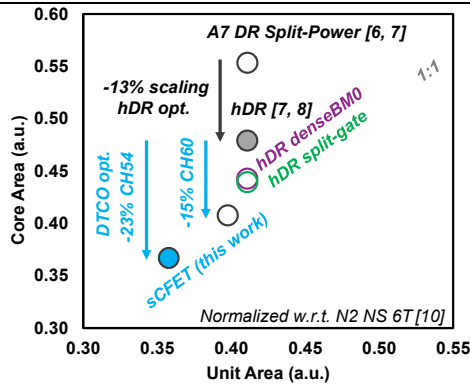


Fig. 3. Core area scaling projections for CFET libraries. hDR optimization scales core area down by 13% than DR. Both hDR denseBM0 and hDR split-gate present a similar scaling potential around 8% on top of hDR. Finally, sCFET reduces core area by 15%-23% (CH60-54) than hDR, as an outcome of denser backside BM0 tracks, split-gate, asymmetric sheet width and vertical local interconnect VMM. sCFET nearly achieves ~1:1 scaling ratio between unit and core area, i.e. sCFET optimizations are converted to core area scaling with high efficiency.

- Full NSH stack with eMDI iso.
- Top NSH patterning stop on eMDI**
- Top Gate patterning
- Top Spacer/InnerSpacer & SD EPI
- Top RMG 1(high-T)**
- Dummy fill, Temporary Bonding, Flip
- Bot Bulk Wafer removal
- Bot NSH patterning stop on eMDI**
- Keep dielectric on to cover channel
- Gate/Gate-Via pre-patterning**
- Bot Gate patterning
- Bot Spacer/InnerSpacer & SD EPI
- Bot WireRelease & Gate/Gate-Via
- Bot RMG (high-T/low-T) incl. plug**
- Bot RMG-Cut
- Bot Contact Trench patterning
- Bot BEOL**
- Bonding/Debonding/Flip
- Top RMG 2(low-T) incl. plug**
- Top RMG-Cut
- Top Contact Trench
- Top BEOL**

Fig. 2. sCFET process flow. List of (FEOL)-modules. Bulk modules shown as gate-aligned cross-sections from process emulation

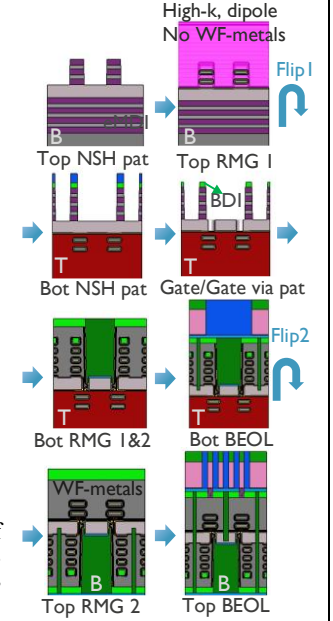


TABLE II. Cell area breakdown (rows $\times$ cols) of CFET libraries. DR shows 26% lower library layout efficiency than N2 6T nanosheet architecture. hDR improves DR layout efficiency from 0.74 to 0.86. Furthermore, hDR denseBM0 enables a higher track density at backside BM0 to improve routability, resulting in low area in many cells, and increased efficiency (0.93). Alternatively, hDR split-gate improves transmission gates, and achieves a similar efficiency (0.94) as hDR denseBM0. Finally, sCFET significantly optimizes cell library, and achieves an overall 98% layout efficiency.

Gate	BM0P (nm)	CH (nm)	CFET Arch	AN2D1	AN2D2	AO22D1	AO22D2	BUFFD1	BUFFD2	DIFFD1	DIFFD2	FASTKD1	ICGD4	INVD1	INVD2	MUX2D1	ND2D1	ND2D2	NR2D1	NR2D2	OA22D1	OA22D2	OR2D1	OR2D2	XNR2D1	XOR2D1	Layout
common	31	62	DR [6,7]	2x3	2x3	2x3	2x3	2x3	2x3	2x11	2x13	2x13	2x11	2x2	2x2	2x6	2x2	2x3	2x2	2x3	2x3	2x3	2x3	2x3	2x3	2x4	0.74
	31	62	hDR [7,8]	1x5	1x5	1x7	1x7	1x7	1x7	1x19	1x12	1x12	1x11	1x2	1x2	1x6	1x3	1x5	1x3	1x3	1x5	1x5	1x5	1x5	1x7	1x7	0.86
	18	62	hDR denseBM0	1x4	1x5	1x7	1x7	1x7	1x7	1x19	1x12	1x12	1x11	1x2	1x2	1x6	1x3	1x5	1x3	1x3	1x5	1x5	1x5	1x5	1x7	1x7	0.93
split	31	62	hDR split-gate	1x4	1x5	1x7	1x7	1x7	1x7	1x19	1x12	1x12	1x11	1x2	1x2	1x6	1x3	1x5	1x3	1x3	1x5	1x5	1x5	1x5	1x7	1x7	0.94
	54-60		sCFET this work	1x4	1x5	1x7	1x7	1x7	1x7	1x19	1x12	1x12	1x11	1x2	1x2	1x6	1x3	1x5	1x3	1x3	1x5	1x5	1x5	1x5	1x7	1x7	0.98
Normalized w.r.t. N2 NS 6T [10]																											
100% Δ area (rows $\times$ cols) 0%																											

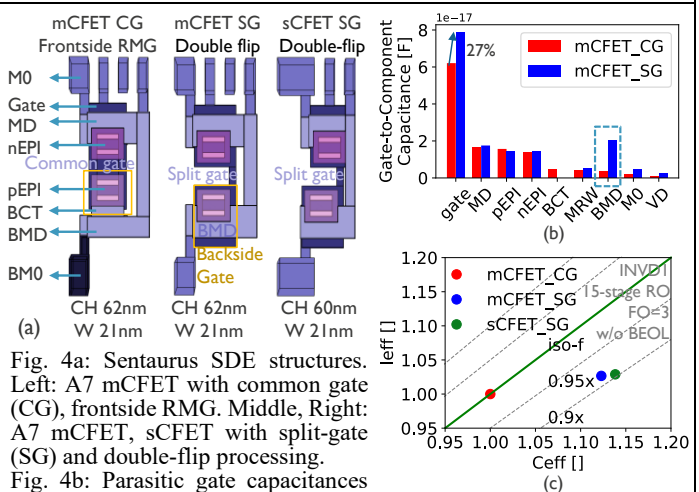


Fig. 4a: Sentaurus SDE structures. Left: A7 mCFET with common gate (CG), frontside RMG. Middle: A7 mCFET, sCFET with split-gate (SG) and double-flip processing. Fig. 4b: Parasitic gate capacitances extracted from mCFET CG and SG using Synopsys Raphael FX. The SG configuration enabled by double-flip results in a 27% increase in gate parasitic capacitance, mainly due to larger gate-to-BMD coupling.

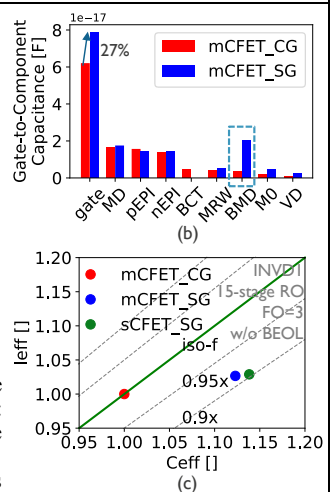


Fig. 4c: RO benchmark [11] results for the 3 CFETs. SG devices show 9-10% lower frequency compared with CG device, attributed to the added capacitance penalty.

Table III: Parameters used in the capacitance sensitivity analysis for gate-over-active and MD/BMD-over-active in sCFET_SG.

DoEs	1	2	3	4	5
Gate over active [nm]	30	25	25	20	10
(B)MD over active [nm]	20	20	15	15	15

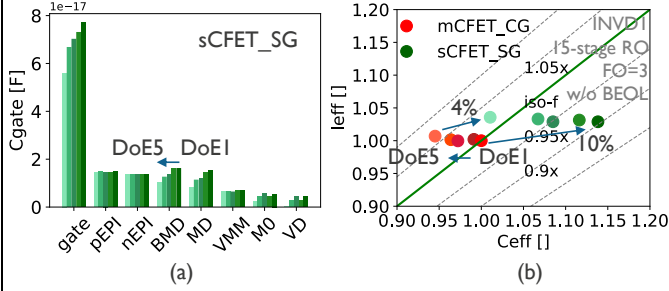


Fig. 5. Sensitivity analysis using the 5 DoEs. (a): Parasitic gate capacitances extracted from sCFET with SG. Reducing the gate and MD/BMD facing area reduces gate parasitic capacitance. (b): Ring oscillator (RO) performance comparison between mCFET_CG and sCFET SG. DoE5 minimizes the frequency gap between mCFET_CG and sCFET_SG from 10% to 4%.

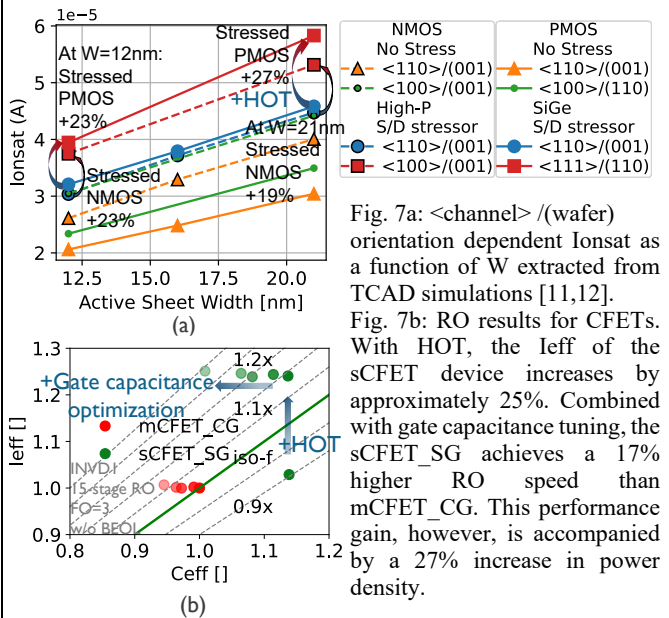


Fig. 7a: <channel>/(wafer) orientation dependent Ionsat as a function of W extracted from TCAD simulations [11,12]. Fig. 7b: RO results for CFETs. With HOT, the leff of the sCFET device increases by approximately 25%. Combined with gate capacitance tuning, the sCFET_SG achieves a 17% higher RO speed than mCFET_CG. This performance gain, however, is accompanied by a 27% increase in power density.

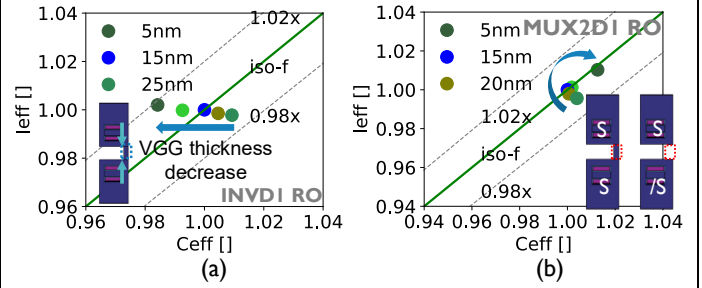


Fig. 6. RO results showing the VGG-thickness sensitivity for split-gate devices, with VGG varied from 5 to 25 nm. (a) INV1 uses only one gate type—a connected gate with uniform polarity—so reducing the VGG thickness lowers the fringing capacitance. (b) In MUX2D1 based RO, the two input terminals of the MUX connected to VSS and VDD, while the select terminal is driven by the RO stage nodes. MUX2D1 includes both connected gates and isolated split-gates with opposite polarities. Thinner VGG improves the connected-gate behavior but increases the capacitive coupling between the isolated opposite-polarity gates. Consequently, the MUX2D1 design exhibits an optimal VGG thickness due to this trade-off.

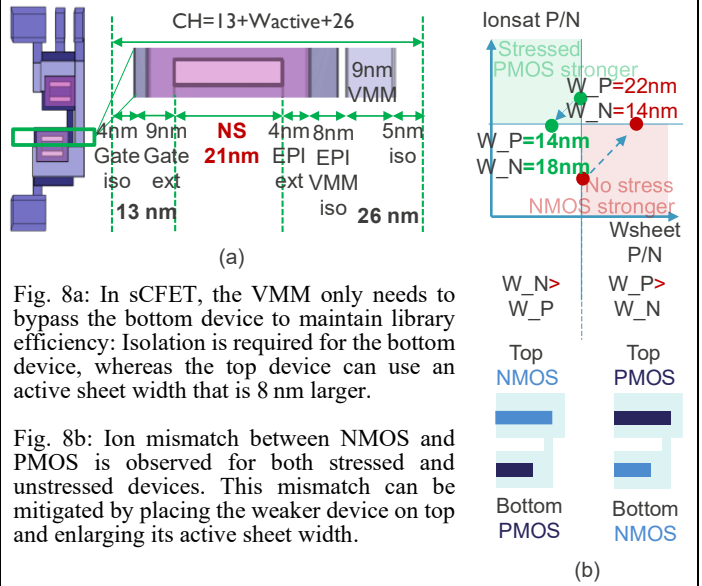


Fig. 8a: In sCFET, the VMM only needs to bypass the bottom device to maintain library efficiency: Isolation is required for the bottom device, whereas the top device can use an active sheet width that is 8 nm larger.

Fig. 8b: Ion mismatch between NMOS and PMOS is observed for both stressed and unstressed devices. This mismatch can be mitigated by placing the weaker device on top and enlarging its active sheet width.

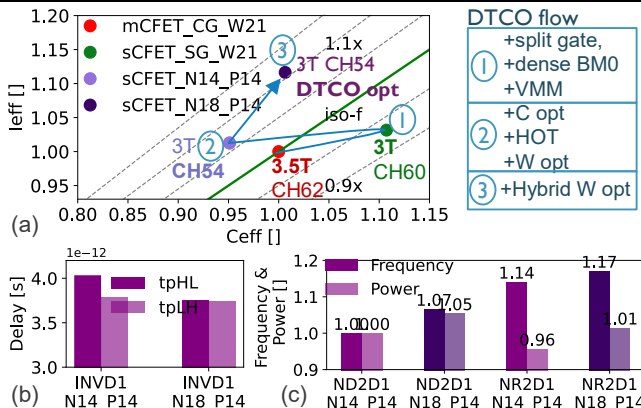


Fig. 9. RO results of CFET devices (DoE4 in Table III). (a): sCFET sheet-width scaling from 21 nm to 14 nm reduces power density while maintaining a 6% higher frequency than mCFET_CG. (b): Increasing NMOS sheet width to 18 nm mitigates rise/fall delay (tpLH, tpHL) imbalance and improves INV1 RO frequency by 5%. (c): NMOS sheet-width enlargement provides larger frequency gains (~7%) in NAND cells, where two NMOS devices are connected in series.

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